



Luca Lazzaroni

Ricercatore a tempo determinato in Elettronica
DITEN – UNIGE

SEDE Genova, Italia

SETTORE RTD-A · IINF-01/A

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BIBLIOMETRIA

SCOPUS

67 DOCUMENTI	338 CITAZIONI	11 INDICE H	20 ARTICOLI, 5 ANNI
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AGG. 5 OTT 2026

01 Profilo

Ricercatore a tempo determinato (RTD-A) presso il Dipartimento di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni (DITEN), Università di Genova, settore IINF-01/A – Elettronica, dal marzo 2025. Dottore di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni dal 2024.

Attività di ricerca su machine learning embedded ed edge AI: toolchain TinyML, reti neurali quantizzate e binarie su microcontrollori e FPGA, percezione e decisione per la guida automatizzata, deep reinforcement learning, LLM on-device. L'attività si svolge anche nell'ambito di progetti di ricerca europei e nazionali con partner industriali.

Edge AI	TinyML	Deep learning embedded	Guida automatizzata	Deep reinforcement learning
LLM on-device				

02 Contatti e profili

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03 Incarichi e formazione

POSIZIONI

Mar 2025 –
oggi

Ricercatore a tempo determinato (RTD-A)
DITEN, Università di Genova
Settore scientifico-disciplinare IINF-01/A — Elettronica.

01/11/2023 – Feb 2025	Assegnista di ricerca DITEN, Università di Genova Assegno: “Realizzazione di modelli di Machine Learning per il trattamento di dati provenienti da veicoli autonomi”.
Nov 2020 – 01/11/2023	Dottorando di ricerca DITEN, Università di Genova – XXXVI ciclo Borsa di dottorato: “Deep learning on the edge”.

FORMAZIONE

2024	Dottorato di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni Università di Genova – conseguito il 14 marzo 2024 Tesi: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”
2020	Laurea Magistrale in Ingegneria Elettronica (LM-29) Università di Genova – 24 luglio 2020 Tesi: “Progetto e implementazione di un motore di esecuzione multiplatforma per applicazioni IoT” · Voto: 110/110
2017	Laurea in Ingegneria Elettronica e Tecnologie dell'Informazione (L-8) Università di Genova – 27 ottobre 2017 Tesi: “Progetto model-based del controllo di robot Lego in ambiente Simulink”
2014	Diploma di Maturità Scientifica Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV) Voto: 100/100

Abilitazione all'esercizio della professione di Ingegnere dell'Informazione — Esame di Stato superato nella seconda sessione 2020.

04 Didattica e incarichi accademici

CORSI

A.A. 2024/25 – 2026/27	Distributed Electronic Systems and Technologies for Environmental Monitoring Docente titolare · cod. 94631
A.A. 2024/25 – 2026/27	Tiny Machine Learning Co-titolare · cod. 114752
A.A. 2026/27	Dispositivi e Circuiti Elettronici Docente titolare · cod. 65997
dall'A.A. 2022/23	Machine Learning for Automated Driving Cultore della materia · cod. 108716

A.A. 2023/24	Computer Graphics Arch. — Games and Simulation Docente a contratto · cod. 172317
A.A. 2022/23	Autonomous Agents in Games Supporto alla didattica · cod. 198216
A.A. 2020/21	Fondamenti di Programmazione di Sistemi Elettronici Supporto alla didattica · cod. 166061
A.A. 2020/21	Approccio Makers alla Progettazione Elettronica Supporto alla didattica · cod. 186985

Tutti i corsi presso l'Università di Genova.

SUPERVISIONE

- 1** Dottorando — Ali Zare, Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni (XLI ciclo)
- 12** Tesi di Laurea — Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)
- 10** Tesi di Laurea Magistrale — Ingegneria Elettronica (LM-29)
- 7** Tesi di Laurea Magistrale — Ingegneria Tecnologica per la Strategia e la Sicurezza (LM/DS)
- 2** Tesi di Laurea Magistrale — Informatica (LM-18)

INCARICHI ACCADEMICI

Membro della Commissione Didattica e Ammissione del Corso di Laurea Magistrale in Engineering for Natural Risk Management.

Responsabile della comunicazione per il Corso di Laurea Magistrale in Electronic Engineering.

05 Curatela e servizio

RUOLI EDITORIALI

2026	Editor del volume APPLEPIES 2026 — Springer LNEE Atti in corso di pubblicazione.
2025	Membro dell'Editorial Board (Early Career) Sensors — MDPI Dal 18 dicembre 2025.
2025	Editor del volume APPLEPIES 2025 — Springer LNEE Atti della International Conference on Applications in Electronics Pervading Industry, Environment and Society.

2024	Guest Editor Energies — MDPI Special Issue “Tiny Machine Learning for Energy Applications”, aprile 2024 — agosto 2025.
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ORGANIZZAZIONE DI CONVEGNI

	Program Chair APPLEPIES 2026 Bologna, Italia, 10–11 settembre 2026.
	Program Chair APPLEPIES 2025 Torino, Italia, 11–12 settembre 2025.
	Program Committee APPLEPIES Dal 2023.
	Program Committee GALA — Games and Learning Alliance Dal 2023.
	Chairman di sessione APPLEPIES 2023 1st Short Oral session — Genova, Italia, 28–29 settembre 2023.

ATTIVITÀ DI REVISIONE

RIVISTE	CONFERENZE
IEEE Access	IEEE ICRA — International Conference on Robotics and Automation
Journal of Systems Architecture	IEEE/RSJ IROS — Intelligent Robots and Systems
Engineering Applications of Artificial Intelligence	GALA — Games and Learning Alliance
Sensors	APPLEPIES
IET Intelligent Transport Systems	
IEEE Transactions on Intelligent Vehicles	
IEEE Transactions on Circuits and Systems II: Express Briefs	
Robotics and Autonomous Systems	
Scientific Reports	
Journal of Big Data	

06 Interventi a convegni

13 INTERVENTI

11-12 set 2025	APPLEPIES 2025 Torino, Italia
11-14 gen 2025	43rd IEEE ICCE — International Conference on Consumer Electronics Las Vegas, NV, USA
19-20 set 2024	APPLEPIES 2024 Torino, Italia
24-26 giu 2024	55th Annual Meeting of the Italian Society of Electronics (SIE) Genova, Italia
28-29 set 2023	APPLEPIES 2023 Genova, Italia
7-9 set 2023	54th Annual Meeting of the Italian Society of Electronics (SIE) Noto (RG), Italia
26-27 set 2022	APPLEPIES 2022 Genova, Italia
7-9 set 2022	53rd Annual Meeting of the Italian Society of Electronics (SIE) Pizzo (VV), Italia
21-22 set 2021	APPLEPIES 2021 Pisa, Italia
1-2 dic 2021	GALA 2021 — Games and Learning Alliance La Spezia, Italia (online)
7-9 lug 2021	52nd Annual Meeting of the Italian Society of Electronics (SIE) Trieste, Italia (online)
9-10 dic 2020	GALA 2020 — Games and Learning Alliance Laval, Francia (online)
19-20 nov 2020	APPLEPIES 2020 Online

07 Premi e associazioni

2025

Best Presentation of the Session — sessione “Edge Computing”

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 gennaio 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

ASSOCIAZIONI PROFESSIONALI

IEEE

IEEE Member e IEEE Young Professional.

IEEE Consumer Technology Society

Membro del Technical Committee “Machine Learning, Deep Learning and AI in CE (MDA)”.

SIE — Società Italiana di Elettronica

Socio nell'Unità di Ricerca dell'Università di Genova dal 2021, area di ricerca “Sistemi elettronici ed applicazioni”.

08 Progetti di ricerca

Synergies

Generation of scenarios for development, training, virtual testing and validation of CCAM systems

Europeo — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework (T8.1)

Genera scenari di test per addestramento, test e validazione di sistemi CCAM, con focus su traffico urbano ed extraurbano — dove la conoscenza degli scenari rilevanti è molto più scarsa che in autostrada. Coinvolge le principali case automobilistiche europee e molte università del settore.

Hi-Drive

Addressing challenges toward the deployment of higher automation

Europeo — H2020, GA 101006664 · Membro dell'unità di ricerca dell'Università di Genova

Sviluppa e testa su larga scala funzionalità di guida automatizzata avanzata, passando dall'automazione condizionata (SAE L3) a quella alta, in diversi scenari e contesti stradali.

L3Pilot

Piloting Automated Driving on European Roads

Europeo — H2020 · Membro dell'unità di ricerca dell'Università di Genova

Sperimentazione su larga scala di funzioni di guida automatizzata SAE L3 su strade pubbliche europee, con le principali case automobilistiche e fornitori europei.

Green Machine Learning for the IoT

Nazionale — PRIN 2022 · Membro dell'unità di ricerca dell'Università di Genova (DII–UNITN / DITEN–UNIGE)

Un framework per applicazioni TinyML pienamente sostenibili distribuite lungo un continuum edge-to-cloud, pensato per dispositivi embedded senza batteria, inclusi quelli privi di connettività di rete.

Assistente Digitale Integrato Automotive

Regionale — POR FESR Piemonte 2014-2020 · Membro dell'unità di ricerca dell'Università di Genova, con Lifetouch

Un assistente virtuale integrato su hardware dedicato a bordo vettura e connesso ai sistemi del veicolo: algoritmi di IA interpretano le richieste del guidatore e agiscono su funzioni veicolo e infotainment.

2026

Cross-platform deployment and characterization of an on-device generative conversational agent

P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonessi, D.P. Pau

Journal of Systems Architecture vol. 181, art. 103981 [Rivista](#)

A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction

L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti

IEEE Signal Processing Letters vol. 33, pp. 136–140 [Rivista](#)

Efficient Certified Robustness Assessment for Transformer Models via Verifier-Aware Perturbation-Position Scheduling

V. Soltanmuradov, R. Berta, **L. Lazzaroni**, D.M. Gómez, A. Pighetti, F. Bellotti

Machine Learning and Knowledge Extraction vol. 8, no. 9, art. 290 [Rivista](#)

FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction

A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti

IEEE Transactions on Circuits and Systems II: Express Briefs vol. 73, no. 8, pp. 888–892 [Rivista](#)

Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms

H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta

IEEE Access vol. 14, pp. 44886–44903 [Rivista](#)

StabilEdge, Formal Verification of Edge AI Applications

L. Lazzaroni, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti

IEEE Access vol. 14, pp. 37416–37434 [Rivista](#)

Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers

A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering pp. 321–327 [Conferenza](#)

Preface

M. Ruó Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero

Lecture Notes in Electrical Engineering pp. v–vi [Editoriale](#)

Optimizing Binary AutoEncoders on STM32 Microcontrollers

H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 301–307

Conferenza

Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students

L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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Conferenza

Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform

H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 218–223

Conferenza

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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Conferenza

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 212–217

Conferenza

Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294

Conferenza

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301

Conferenza

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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Embedded Deployment of an LLM-Based Voice Assistant

L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 273–278

Conferenza

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 286–291

Conferenza

A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 183–190

Conferenza

A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 249–254

Conferenza

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 79–84

Conferenza

2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874

Rivista

TinyML Acceleration with MAX78000

A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474

Conferenza

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 156–163

Conferenza

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking

H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonhi, V. Hashemi, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 387–394

Conferenza

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonhi, V. Hashemi, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 139–147

Conferenza

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset

R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 395–402

Conferenza

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

IEEE Sensors Letters vol. 9, no. 9, art. 6010204

Rivista

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 497–504

Conferenza

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board

H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–249

Conferenza

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection

M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta

IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369

Rivista

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices

A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 323–330

Conferenza

Green Machine Learning for the IoT (GEMINI): A Position Paper

L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 339–346

Conferenza

Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 123–130

Conferenza

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta

IEEE Access vol. 13, pp. 26589–26607

Rivista

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 250–257

Conferenza

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti

2025 IEEE International Conference on Consumer Electronics (ICCE)

Digest of Technical Papers IEEE International Conference on Consumer Electronics

Conferenza

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 234–241

Conferenza

2024

An embedded end-to-end voice assistant

L. Lazzaroni, F. Bellotti, R. Berta

Engineering Applications of Artificial Intelligence vol. 136, art. 108998

Rivista

Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving

Workshop

CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti

Electronics vol. 13, no. 9, art. 1624

Rivista

End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta

Electronics vol. 13, no. 7, art. 1286

Rivista

YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 418–423

Conferenza

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 444–449

Conferenza

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 124–129

Conferenza

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244

Rivista

Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960

Rivista

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta

Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356

Conferenza

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794

Rivista

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 352–357

Conferenza

Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 170–180

Conferenza

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta

International Journal of Serious Games vol. 10, no. 4, pp. 153–170

Rivista

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion

Applied Sciences vol. 13, no. 11, art. 6432

Rivista

Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta

2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)

IEEE International Symposium on Industrial Electronics

Conferenza

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 307–311

Conferenza

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

IEEE Access vol. 11, pp. 28522–28550

Rivista

Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 9–15

Conferenza

Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

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Deep Reinforcement Learning for Automated Car Parking

L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

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Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

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Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**

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An Embedded End-to-End Voice Assistant

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Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora

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Classifying Simulated Driving Scenarios from Automated Cars

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A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

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Project and Development of an Italian Offline Embedded Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste

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Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

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Employing an IoT Framework as a Generic Serious Games Analytics Engine

L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

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